Scope of Work

1. Predictive Demand Forecasting: From Reactive to Proactive

Transition from SO-triggered production to probability-triggered production planning.

Requirements:

- **Time-Series Forecasting Models**

  Use advanced models (e.g., Prophet, LSTM) to identify:

  - Seasonal demand trends
  - Customer-specific purchase cycles
  - Alloy-wise and diameter-wise demand patterns
- **CRM Data Ingestion**

  - Integrate CRM lead status and project probability
  - Assign weighted probabilities to expected orders
  - Convert pipeline visibility into billet and alloy forecast
- **Output**

  Rolling 15-30-45-day demand forecast:

  - Alloy-wise
  - Diameter-wise
  - Daily refresh capability based on orders punched
  - Direct integration with procurement and foundry planning

2. Dynamic Billet Planning & Furnace Optimization

AI to function as a “Digital Foreman” for constraint-based scheduling.

Requirements:

- **Constraint-Based Scheduling Engine**

  - Map furnace capacities and alloy production batch cycles
  - Diameter changeover penalties
  - Alloy sequencing constraints
  - Maintenance downtime integration
- **Intelligent Order Grouping**

  - Batch similar alloys and diameters
  - Optimize heat efficiency
  - Minimize dross loss
  - Reduce downtime from die and diameter change
- **Dynamic Safety Stock Logic**

  Replace static safety stock with:

  - Lead-time-based dynamic buffers
  - Port/shipping delay intelligence

        - Supplier reliability scoring

3. “Smart Charge” Mix – Linear Programming (LP Engine)

AI-driven charge sheet optimization for lowest cost to produce (LCP).

Requirements:

- **Linear Programming Optimization**
    - Minimize cost of melt charge
    - Consider:
        - LME-linked primary ingot pricing
        - Internal scrap
        - External scrap grades (Tense, Tabor, etc.)
        - Master alloys
- **Chemical Constraint Modelling**
    - Respect alloy ceilings (Fe, Si, Mg, Mn, etc.)
    - Optimize chemical headroom utilization
    - Maximize use of low-cost scrap within spectrometer compliance
- **Real-Time Recipe Adjustment**
    - Daily scrap price fluctuation sensitivity
    - Dynamic furnace-wise charge sheet recommendation, based on live sampling

4. Procurement Intelligence & Strategic Arbitrage

Upgrade procurement into a market-driven intelligence function.

Requirements:

- **Supplier Lead-Time Intelligence**
    - Track historical delivery performance
    - Adjust ROP dynamically
    - Predict supply risk
    -
- **Market Spread Monitoring**
    - Monitor LME vs Scrap spreads
    - Trigger “Strategic Buy” signals based on:
        - 15-30-45 day forecast alignment
        - Market discount windows
        -
- **Additive & Master Alloy Optimisation**
    - Track alloy mix vs forecast
    - Prevent emergency procurement
    - Optimize inventory turnover

5. Customer Health & Profitability Matrix

Move from gross margin analysis to true Net Contribution intelligence.

A. Scoring Framework

The system must calculate customer score incorporating:

Financial Metrics

- Credit carrying cost
- Delayed payment impact
- Finished goods holding cost
- Quality debit notes %

Operational Metrics

- Die amortization vs volume
- Average Attempts to Complete (AtC)
- Die reset penalties
- Recovery rates (theoretical vs actual)
- Complexity vs margin
- Delayed lifting of finished goods

B. Customer Segmentation Matrix

Category	Characteristics	Recommended Action
Star Partners	High volume, high recovery, on-time payment	Protect & Prioritize
Technical Specialists	High complexity, high margin	Price for complexity
Volume Fillers	Low margin, high recovery, fast payment	Maintain
Resource Drainers	Low margin, high resets, slow payment	Correct / Reprice / Exit

C. AI-Driven Corrective Triggers

The system must automatically flag:

- Debit Notes > 2% of revenue
- Avg Die Loads > 1.5
- Margin below threshold for 3 consecutive cycles
- Credit exposure above limit

And suggest:

- Complexity surcharge
- Price revision
- Volume commitment requirement
- Relationship termination review

Deliverables Expected from Consultant

1. End-to-end AI architecture design
2. Data pipeline integration (ERP + CRM + Production + LME feed)
3. Power BI dashboards for:
    - Forecasting
    - RM & Billet inventory optimisation
    - Melt optimisation
    - Procurement advisory
    - Customer scoring & segmentation
4. Pilot implementation:
    - 2 alloys
    - 30 customers
5. Full-scale deployment roadmap
6. Training of internal analytics & planning team
7. Performance-linked improvement framework

Key KPI Targets

The proposed system must target measurable improvements:

- 3–5% reduction in metal cost per MT
- 10–15% reduction in billet & raw material inventory
- 5–10% improvement in furnace utilization
- 10–15 day reduction in working capital cycle
- Identification and corrective action on bottom 15% value-destructive customers